HYDRODYNAMIC SINGULARITIES

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ABSTRACT

Hydrodynamic singularities appear when a smooth free surface focuses motion into a neck, bridge, rim, or tip. The global flow may be millimetres wide, but the decisive balance can be set by a much smaller inner region.

Poster content placeholder: replace with the final student-facing story once the figures are chosen.

HIGHLIGHTS

Key questions

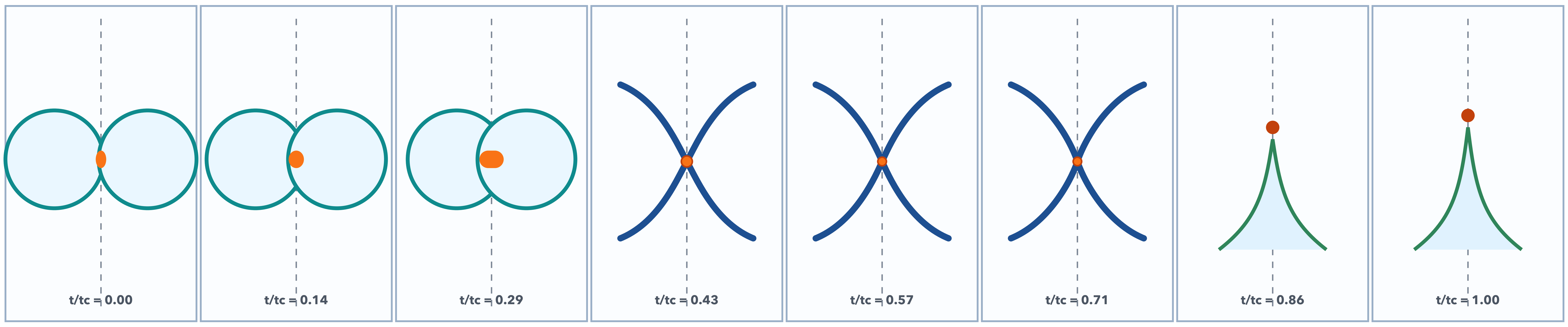
- What local length scale is collapsing?

- Which balance resolves the near-singular region?
- How does a small neck or tip control the whole flow?

Useful numbers: Oh, We, Ca, De, Wi

FROM SMOOTH MOTION TO A TINY DECISIVE REGION

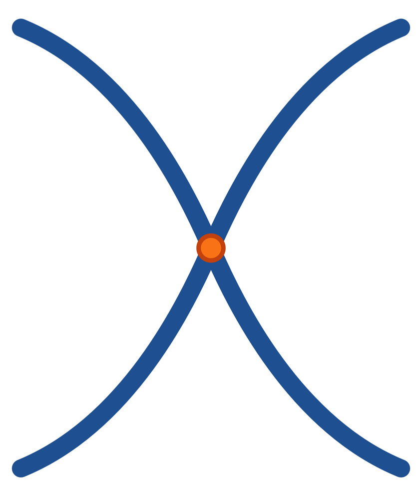
One visual spine: coalescence, necking, jetting, and breakup



Use this central strip for real Basilisk or experimental frames later. Keep time labels and one colour field; do not bury the story in tiny subpanels.

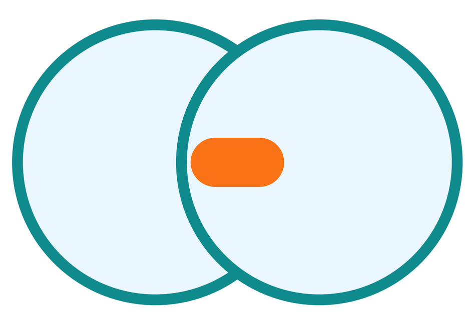
LOCAL GEOMETRY

Pinch-off



The minimum radius becomes the natural clock. Competing balances decide the thinning route.

Coalescence



A microscopic bridge reshapes both drops, coupling local curvature to global motion.

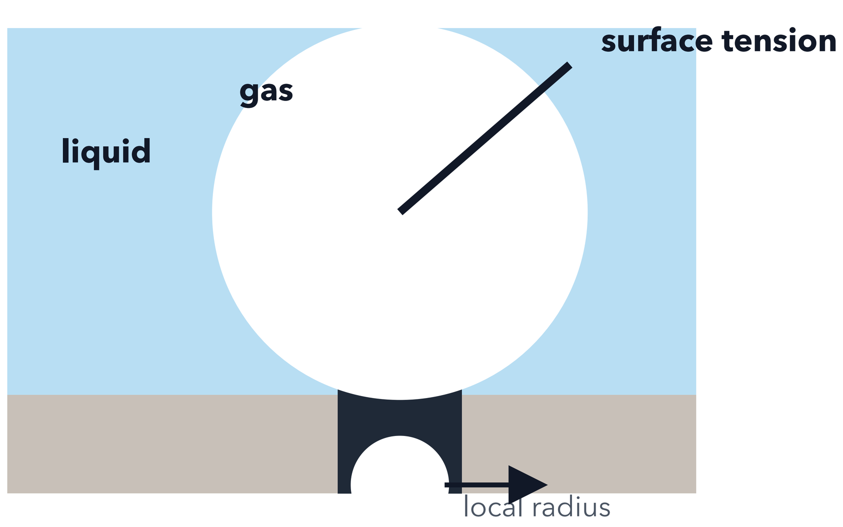
PATHWAYS

Jets and sheets

Rims, tips, ligaments, and droplets are different routes to the same question: where does the flow focus next?

WHAT SETS THE CUTOFF?

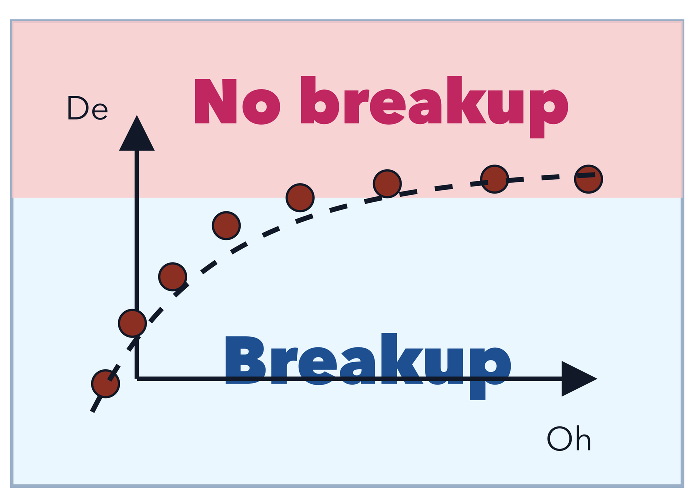
Placeholder schematic



Replace with the final mechanism drawing: neck radius, outer scale, stress balance, and the dimensionless groups that matter.

CONCLUSION

Regime-map slot



Use this for the final take-home message: which regime breaks, which regime survives, and what physics changes the route.

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